Non–Archimedean Geometry

Based on a lecture series by Enrica Mazzon.

December 2024

Introduction to non–Archimedean geometry and applications of the techniques to algebraic geometry.

§0.1 About. This course on pseudolattices was a part of a Winter School for students and early career researchers, organised by the UK Algebraic Geometry Network. The lectures were delivered in-person in the University of Warwick, and were transcribed by Parth Shimpi. These notes have undergone several amendments and are not a verbatim recall of the lectures, therefore discretion is advised when using this material. They are available online at `https://pas201.user.srcf.net/documents/2023-ukag-pseudolattices.pdf`. All errors and corrections should be communicated to by email to `parth.shimpi@glasgow.ac.uk`.

§1 Non–Archimedean fields

Let $\mathbb{F}$ be a field. A *semi-norm* is a map $|\cdot| : \mathbb{F} \to \mathbb{R}_{\geqslant 0}$ such that $|0| = 0$, $|1| = 1$, $|a+b| \leqslant |a|+|b|$, and $|ab| \leqslant |a|\cdot|b|$. If moreover we require that $|ab| = |a| \cdot |b|$, then the seminorm is *multiplicative*. If $|a| > 0$ whenever $a \neq 0$, then $|\cdot|$ is called a *norm*.

A semi-norm $|\cdot|$ is non–Archimedean if it satisfies the strong triangle inequality $|a + b| \leqslant \max\{|a|, |b|\}$.

Remark 1.1. If $|\cdot| : \mathbb{F} \to \mathbb{R}_{\geqslant 0}$ is a non–Archimedean multiplicative norm, then $v(-) = -\log|\cdot|$ is a non–Archimedean valuation.

We say $(K, |\cdot|)$ is a non–Archimedean field if it is a field endowed with a non–Archimedean multiplicative norm. In this case, we say $R = K^0 = \{x \in K \mid |x| \leqslant 1\}$ is the *valuation ring*, $K^{\infty} = \mathfrak{m} = \{x \in K \mid |x| < 1\}$ is the maximal ideal, and $\tilde{K} = R/\mathfrak{m}$ is the residue field of K.

Example 1.2. Any field $\mathbb{F}$ can be endowed with the trivial norm $|\cdot|_0$ defined as $|0|_0 = 0, |a|_0 = 1$. Then $\mathbb{F} = \mathbb{F}^0 = \tilde{\mathbb{F}}$ and $\mathbb{F}^{\infty} = \{0\}$.

Example 1.3. If $\mathbb{F}$ is any field, then we can consider the associated field of Laurent series

$$K = \mathbb{F}((t)) = \{\sum_{i=N}^{\infty} a_i t^i \text{ for some } N \in \mathbb{Z}, a_i \in \mathbb{F}\}.$$

Then any $\varepsilon \in (0, 1)$ determines a norm $|0| = 0$, $|f| = \varepsilon^{\mathrm{ord}_t(f)}$. Then $K^0 = \mathbb{F}[\![t]\!]$, $K^{\infty} = (t)$, $\tilde{K} = \mathbb{F}$.

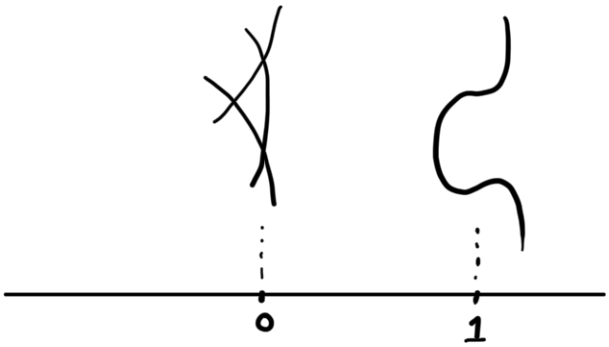
Example 1.4. The field of p-adic rationals

$$\mathbb{Q}_p = \{\sum_{i=N}^{\infty} a_i p^i \text{ for some } N \in \mathbb{Z}, a_i \in \{0, ..., p-1\}\}.$$

Again, any $\varepsilon \in (0, 1)$ determines a norm $|s| = p^{-\mathrm{ord}_p(s)}$. It is customary to take $\varepsilon = \frac{1}{p}$ in number theory. Have $\mathbb{Q}_p^0 = \mathbb{Z}_p$, $\mathbb{Q}_p^{\infty} = (p)$, $\widetilde{\mathbb{Q}_p} = \mathbb{F}_p$.

Remark 1.5. A norm $|\cdot|$ is Archimedean if it satisfies the Archimedean property, i.e. for every $r \in \mathbb{R}_{\geqslant 0}$ there exists an $n \in \mathbb{Z}$ with $|n| \geqslant r$. Show that a norm is non–Archimedean if and only if $|\cdot|$ is uniformly defined on the image of $\mathbb{Z} \to K$.

§1.1 Motivation for non–Archimedean fields in algebraic geometry. Say you wish to understand the geometry of a smooth projective $\mathbb{C}$–variety X. A possible technique is to place X in a 1–parameter family $f : \mathcal{X} \to \mathbb{A}^1 = \mathrm{Spec}(\mathbb{C}[t])$ i.e. $\mathcal{X}_1 = f^{-1}(1) \cong X$ and $\mathcal{X}_0 = f^{-1}(0)$ (a *degeneration*) is a union of irreducible components with simpler structure, thus turning some of the geometric complexity combinatorial.



Then we consider the base change of $\mathcal{X}$ to a infinitesimal punctured tubular neighbourhood of $\mathcal{X}_0$, to obtain $\mathcal{X} \times_{\mathbb{C}[t]} \mathrm{Spec}(\mathbb{C}((t)))$. This is a variety over $K = \mathbb{C}((t))$.

K–analytic manifolds. Over $\mathbb{C}$ with usual norm, a smooth algebraic variety X is locally defined by polynomials $f_1, ..., f_m \in \mathbb{C}[x_1, ..., x_n]$. One can turn X into a complex manifold $X(\mathbb{C})$, by locally considering the subset $\{(x_1, ..., x_n) \in \mathbb{C}^n \mid f_i(x_1, ..., x_n) = 0\}$ with the geometry inherited from $\mathbb{C}^n$. This allows us to apply new techniques (complex analysis, differential geometry, Hodge theory, etc) to study X. When X is proper, this process of analytification incurs no loss of information in the sense of Serre's GAGA theorem– two smooth proper $\mathbb{C}$–varieties X, Y are isomorphic if and only if $X(\mathbb{C}) \cong Y(\mathbb{C})$ as complex manifolds.

What's the topology on $X(\mathbb{C})$? It is the weakest topology such that for any Zariski–open set $U \subset X$, the subset $U(\mathbb{C}) \subset X(\mathbb{C})$ is open and for any regular function $f \in \mathcal{O}_X(U)$ the map $|f| : U(\mathbb{C}) \to \mathbb{R}$ is continuous.

What's the sheaf of functions? We consider *analytic functions*, on an open set $V \subset X(\mathbb{C})$ these are those that can be written locally as a uniform limit of regular functions. Then $X^{an} = (X(\mathbb{C}), \mathcal{O}_{X(\mathbb{C})})$ is a complex manifold.

When K is a non–Archimedean field, the naïve idea is to replicate the same construction. However, the analogue of the GAGA principle fails in a spectacular way. To see this, given $a \in K, r \in \mathbb{R}_{\geqslant 0}$ write $B^+(a, r), B^-(a, r)$ for the closed and open balls of radius r centered at a respectively. In the trivially valued case, $B^+(a, 1) = K$, $B^+(a, \frac{1}{2}) = a$. In particular the metric topology on K is discrete. In the non–trivially valued case, consider $r' \leqslant r$ and suppose $B^+(0, r) \cap B^+(a, r')$ is non–empty. Then in fact $B^+(a, r') \subseteq B^+(0, r)$, i.e. closed balls are either disjoint or nested. Exercise: every point in a ball is a center of the ball. Exercise: every ball in K is both open and closed in the metric topology. Thus the metric topology is totally disconnected, i.e. the only connected subsets have cardinality $\leqslant 1$.

Issues: the characteristic (indicator) function of a ball is continous and analytic. There are too many analytic functions, and thus too many analytic isomorphisms. Thus have the 'death of GAGA': Serre proved every n–dimensional $\mathbb{Q}_p$–analytic manifold in this naïve sense is isomorphic to a disjoint union of up to $p - 1$ copies of $B^+(0, 1)^n$.

§ 2 Berkovich spaces

In the 90s, Berkovich adopted the philosophy that the topology of the K–analytification is itself not a problem; rather there are many points that are missing from the definition of a K–analytic manifold. The resulting theory becomes very close to the that of $\mathbb{R}$ and $\mathbb{C}$–manifolds.

Definition 2.1. Let K be a complete non–archimedean field, and X a K–variety. We set

$$X^{an} = \{x \in (\zeta_x, |\cdot|_x) \mid \zeta_x \text{ is a point of } X, \quad |\cdot|_x \text{ is a multiplicative norm on the residue field } k(\zeta_x) \text{ extending } |\cdot| \text{ on } K\}$$

Remark 2.2. If ζ_x is a closed point then $k(\zeta_x)$ is a finite extension of K and thus there is a unique $|\cdot|_x$. Thus $X(K)$ is a subset of X^{an}.

If ζ_x is not a closed point then the extension $k(\zeta_x)/K$ is transcendental and thus there are infinitely many ways to extend the seminorm. This forms the 'main' part of X^{an}.

We endow X^{an} with the weakest topology such that the forgetful map $i : X^{an} \to X$ is continuous, and for all open $U \subset X$ and for all $f \in \mathcal{O}_X(U)$, the map $i^{-1}(U) \to \mathbb{R}$ given by $(\zeta_x, |\cdot|_x) \mapsto |f(\zeta_x)|_x$ is continuous.

The sheaf of analytic functions comes by assigning every open set $V \subset X^{an}$ to the set of functions thhat are locally a uniform limit of regular functions.

§2.1 Properties.

0. Taking $K = \mathbb{C}$ with the usual norm gives the usual complex manifold $X(\mathbb{C})$– this is because the only valued extension of $\mathbb{C}$ is trivial so the non-closed points are not a part of X^{an}.
1. X^{an} is locally compact and locally path connected. Moreover, X is connected (separated, proper) if and only if X^{an} is connected (resp. Hausdorff, compact).

   (Howshovski–Loeser) show that X^{an} is locally contractible and has the homotopy type of a polyhedron.
2. GAGA-type results hold, and there is an equivalence between proper algebraic K-varieties and their analytifications.

Example 2.3. If $X = \operatorname{Spec} A$ is affine for some K-algebra A, then the set X^{an} can be described as the set of all multiplicative seminorms $|\cdot|_x : A \to \mathbb{R}_{\geqslant 0}$ extending $|\cdot|$. The topology is then the weakest such that $|f| : X^{an} \to \mathbb{R}$ is continuous for all $f \in A$.

Example 2.4 (The Berkovich line). Consider $X = \mathbb{A}^1_K = \operatorname{Spec} K[y]$, and suppose K is an algebraically closed complete non-Archimedean field.

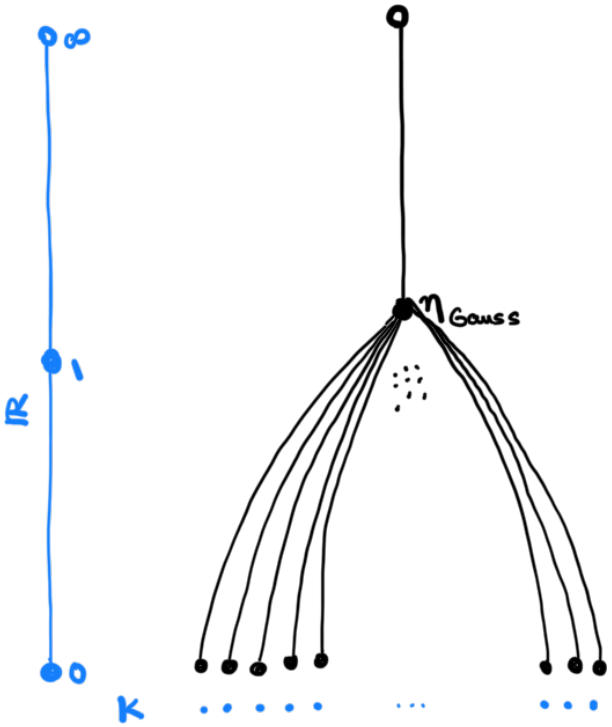
If K is equipped with the trivial norm $|\cdot|_0$, then a point $x \in X^{an}$ corresponds to multiplicative seminorms $|\cdot|_x : k[y] \to \mathbb{R}_{\geqslant 0}$ extending the trivial norm. Since K is algebraically closed, it suffices to describe such a seminorm on linear polynomials of the form $(y-a)$. One such norm is the 'Gauss norm' η_{Gauss}, which is simply the trivial norm $\eta_{\text{Gauss}}(y-a) = 1$ for all $a \in K$.

Every $a \in K$ defines a seminorm $\eta_{a,0} : K[y] \to \mathbb{R}$ given by $f \mapsto |f(a)|_0$, i.e. $\eta_{a,0}(y-b) = |a-b|_0 = 1$ if $b \neq a$ and $\eta_{a,0}(y-a) = 0$.

On the other hand, every $a \in K$ and $r \in (0,1)$ defines a seminorm $\eta_{a,r}$ given by $f \mapsto r^{\operatorname{ord}_a(f)}$. Thus $\eta_{a,r}(y-a) = r$ and $\eta_{a,r}(y-b) = 1$ if $b \neq a$. In the limit $r \to 0$ this recovers $\eta_{a,0}$, and in the limit $r \to 1$ this recovers η_{Gauss}.

For $r \in (1,\infty)$, we can also define the seminorm $\eta_{\infty,r}$ given by $f \mapsto r^{\deg(f)}$. In the limit $r \to 1$ this again recovers η_{Gauss}.

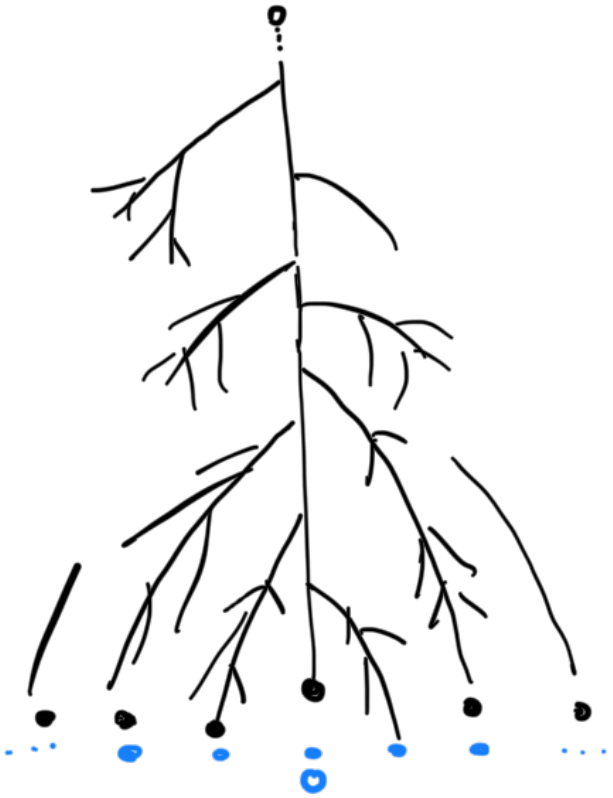
These are all the points of X^{an} which can be pictorially represented as follows.



To describe the topology, consider the polynomial $f = y$. The open sets determined by this are $U_{s<s'} = \{x \in X^{an} \mid s < |y|_x$ for $s < s'$ in $\mathbb{R}$. If $s' < 1$ then this is $\bigcup_{a \in K}\{\eta_{a,r} \mid r \in (s,s')\}$. If $s > 1$ then $\{\eta_{\infty,r} \mid r \in (s,s')\}$. If $s < 1$ and $s' > 1$ then ...

Thus the topology is the metric topology on legs $[0,1)$ and $(1,\infty)$, while any open neighbourhood of η_{Gauss} contains all but finitely many legs.

If K is algebraically closed and non-trivially valued The points $\eta_{a,0} : f \mapsto |f(a)|$ can still be defined. On the other hand for $a \in K$ and $r > 0$, we can define $\eta_{a,r}$ by $f \mapsto \sup_{z \in B^+(a,r)} |f(z)|$. If $r \geqslant |a|_K$, then $B^+(a,r) = B^+(0,r)$ and so $\eta_{a,r} = \eta_{0,r}$ for r sufficiently large. More generally, $\eta_{b,r} = \eta_{c,r}$ for $r \geqslant |b-c|$. This gives an infinite tree branching at every value of K. The points correspond to open balls in K or sequences of nested balls. It is possible for an infinite intersection of nested balls to be empty.



The topology is such that any open neighbourhood of a branching point contains all but finitely many branches.

If $K, |\cdot|$ is a non-Archimedean field, then taking $-\log|\cdot|$ gives a non-Archimedean *valuation* $v : K \to \overline{\mathbb{R}} = \mathbb{R} \cup \{\infty\}$, i.e. a valuation such that $v(a+b) \geqslant \min\{v(a), v(b)\}$.

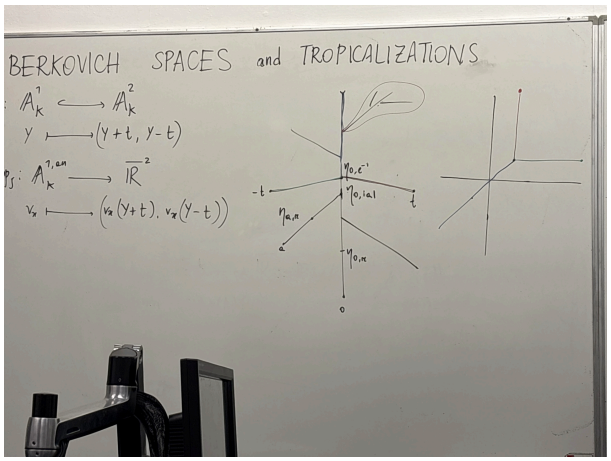
Let K be algebraically closed and non-trivially valued. The *tropicalisation map* of $\mathbb{A}^n_K$ is defined as $\mathrm{trop} : \mathbb{A}^n_K(K) \to \overline{\mathbb{R}}^n$ given by $(x_1, ..., x_n) \mapsto (v(x_1), ..., v(x_n))$, and extend it to the Birkovich analytification by $((y_1, ..., y_n), v_x) \mapsto (v_x(y_1), ...,$

The *tropicalisation* of $\mathbb{A}^n_K$ is the closure of $\mathrm{trop}(\mathbb{A}^n_K(K))$ which is equal to $\mathrm{trop}(\mathbb{A}^{n\,an}_K)$.

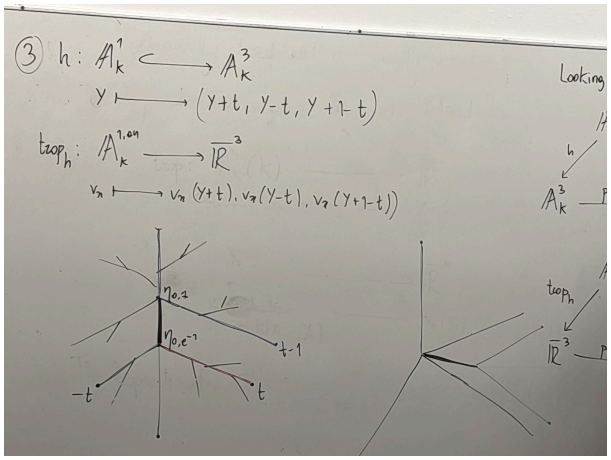
For $X \subseteq \mathbb{A}^n_K$, can define tropicalisation by restriction, but this actually depends on the embedding of X into $\mathbb{A}^n_K$.

Example 2.5. Fix $X = \mathbb{A}^1_K = \operatorname{Spec} K[y]$.

1. if we take the trivial inclusion $\mathrm{id} : \mathbb{A}^1_K \to \mathbb{A}^1_K$, then the tropicalisation with respect to this inclusion sends a seminorm $\eta \in \mathbb{A}^{1\,an}_K$ to $-\log \eta(y)$. Thus $\mathrm{trop}(\eta_{0,0}) = \infty$, $\mathrm{trop}(\eta_{0,r}) = -\log(r)$, while $\mathrm{trop}(\eta(a,r)) = -\log|a|$ whenever $r < |a|$. Every branch gets 'contracted back to branch point'.
2. On the other hand consider the inclusion $\mathbb{A}^1_K \hookrightarrow \mathbb{A}^2_K$ by $y \mapsto (y+t, y-t)$, so that η maps to $(\log \eta(y+t), \log \eta(y-t))$.



3. If we take the inclusion $(y+t, y-t, y+1-t)$ in $\mathbb{A}^3$, get similar picture, with two trivalent vertices and four infinite arms.



Observe that tropicalisation 3 projects onto tropicalisation 2.

Theorem 2.6 (Payne). *For K algebraically closed and non-trivially valued, with X a closed subvariety of $\mathbb{A}^n/\mathbb{P}^n/$some torc variety, taking the inverse limit of Trop(X,i) over all inclusions i is homeomorphic to X^{an}.*

§ 3 Dual complexes

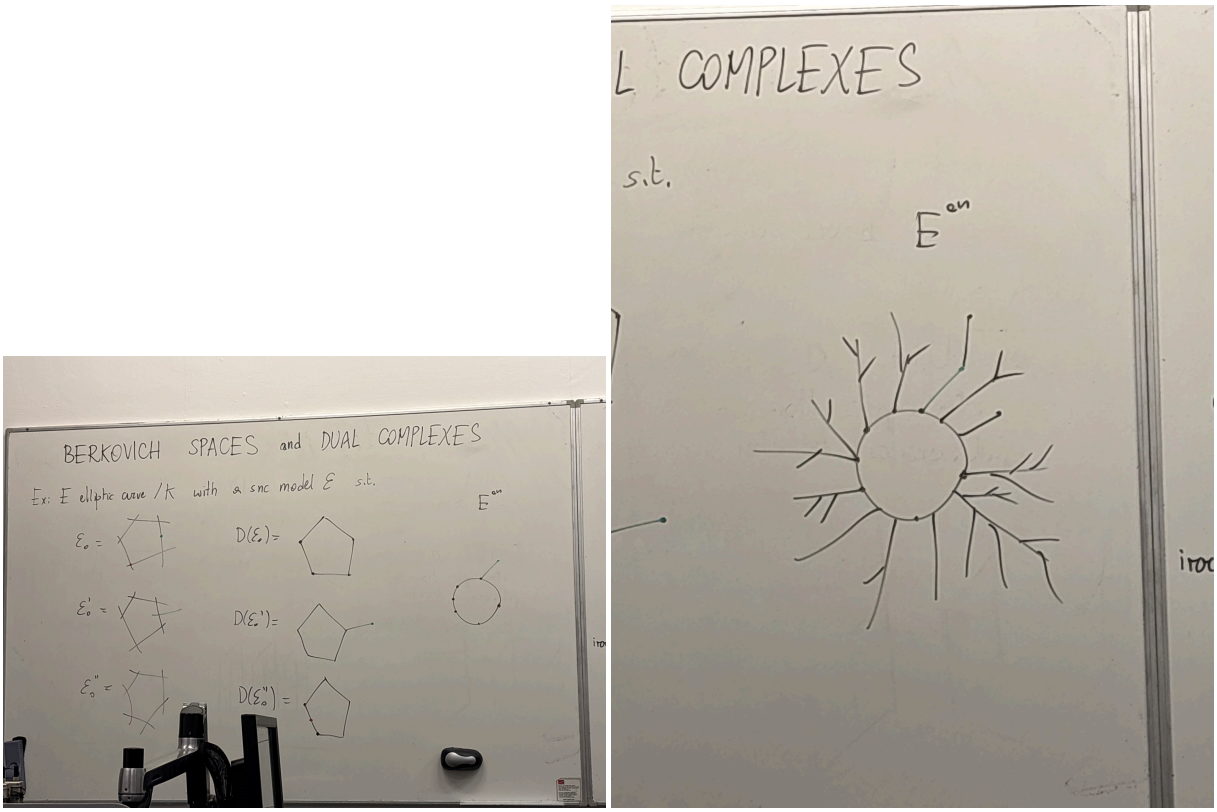
Consider the non-Archimedean field $K = \mathbb{C}(\!(t)\!)$, then a K-variety is a family of varieties parametrised over $t \neq 0$ [IMAGE OF CONIC FAMILY XY=T]

Writing $R = \mathbb{C}[\![t]\!]$ for the valuation ring, we say a degeneration of the family over $t = 0$ is an snc model $\mathcal{X}$ of X over R where we require the special fiber $\mathcal{X}_0$ to be a snc divisor with smooth components. To this we associate a *dual complex* where each component of $\mathcal{X}_0$ is a vertex, and there is a k-simplex $[D_1, D_2, ..., D_k]$ for every irreducible non-empty component of $D_1 \cap D_2 \cap ... \cap D_k$.

The dual complex of $\mathcal{X}_0$ embeds into X^{an}, i.e. for every point of the dual complex we can associate a valuation of $K[x,y]/(xy-t)$. The point associated to the point $\alpha \in D_1 \bullet - \bullet D_2 \cong [0,1]$ corresponds to valuation $v_\alpha(x) = 1-\alpha, v_\alpha(y) = \alpha$.

Proposition 3.1. *Let X be a smooth proper vareity over $K = \mathbb{C}(\!(t)\!)$. Then for any snc model $\mathcal{X}$ of X over $R = \mathbb{C}[\![t]\!]$, the dual complex of $\mathcal{X}_0$ embeds into X^{an}. We say the image $Sk(\mathcal{X})$ is the skeleton. Further, X^{an} retracts onto this skeleton and this is actualy a strong deformation retraction.*

Taking the inverse limit over all snc models recovers the Birkovich analytification X^{an}.



Visual picture of how retraction works.

§ 4 Smooth projective CY setting

Let X be a smooth projective CYn over $K = \mathbb{C}(\!(t)\!)$. Also assume there exists a family $Y \to C$ over a smooth complex curve C, and X is the base change of Y to $\mathbb{C}(\!(t)\!)$ a punctured neighbourhood of some $c \in C$. This is necessary for some arguments which require Y to be of finite type.

Can we pin down an snc model $(\mathcal{X}, \mathcal{X}_0)$ such that the skeleton of $\mathcal{X}_0$ captures the essential geometry (i.e. every other snc skeleton retracts? For elliptic curve this is pentagon with leg sticking out for some reason).

Run MMP on $(\mathcal{X}, \mathcal{X}_0)$ to get a minimal dlt pair $\mathcal{Y}, \mathcal{Y}_0$. it is a fact that X has a projective good minimal dlt model over R, where we require that $(\mathcal{Y}, \mathcal{Y}_{0,\mathrm{red}}$ is dlt. DLT ensures that the dual complex of $\mathcal{Y}_0$ can still be defined, it is the dual complex of reduced $\mathcal{Y}_0$.

minimal– $\mathcal{Y}_{0,\mathrm{red}}$ is torsion.

good– components of $\mathcal{Y}_{0,\mathrm{red}}$ are $\mathbb{Q}$–cartier.

[deFernex–Kollar–Xu] study behaviour of dual complex under MMP, and show that $D(\mathcal{Y}_0)$ is obtained from $D(\mathcal{X}_0)$ using a sequence of collapses.

[Kollar–Mori, deFernex–Kollar–Xu] The PL homeomorphism type of $D(\mathcal{Y}_0)$ is independent of choice of minimal dlt model $\mathcal{Y}$. Dual complexes of crepant birational models are PL homeomorphic.

Definition 4.1. The essential skeleton Sk(X) is the image in X^{an} of the dual complex of any minimal dlt model.

Remarks– you can take any snc model and any minimal dlt to define the skeleton. The skeleton of a minimal DLT model sits inside the skeleton of any snc model. "Which valuations inside X^{an} are essential?"

Alternatively [Kontsevich–Soibelman, Mustata–Nicaise] it is a union of essential faces of $Sk(\mathcal{X})$ or X^{an} corresponding to strata where a canonical form ωhas minimal weight (discrepancy) in a suitable sense.

If haev a good minimal dlt model $\mathcal{Y}$, then there is a retraction map $\rho_{\mathcal{Y}} : X^{an} \to Sk(X)$ that depends on $\mathcal{Y}$. This is a strong deformation retraction. In particular X^{an} and Sk(X) have the same homotopy type.

Proposition (Nicaise–Xu) Sk(X) is a pseudomanifold with boundary, i.e. there exists a triangulation in which Sk(X) is a union of equi-dimensional m-simplices, every $(m-1)$–simplex is a face of at most two m–simplices, and any two m–simplices are connected by simple wall crossing (strong connectedness).

Now consider X a maximally degenerate (i.e. Sk(X) is dim(X)-dimensional) smooth projective CYn. If $n = 1$ then the essential skeleton is S^1. If $n = 2$ then it is $S^2 \cong \mathbb{CP}^1$ by [Kulikov]. If X is strict Calabi-Yau (i.e. $h^{i,0}(X) = 0$ for all $0 < i < n$) then Nicaise-Xu show that Sk(X) has $\mathbb{Q}$-homology of S^n. Dont know if it is a sphere except in specific cases.

If X is hyperKähler, then Sk(X) has rational homology of $\mathbb{CP}^{n/2}$ by [Jollar-Laza-Sacca-Voisin]. Dont know if it is a projective space except in specific cases.

§ 4.1 Connection with mirror symmetry. Kotsevich-Soibelman think that the retraction $X^{an} \to Sk(X)$ is the non-archimedean analog of an SYZ fibration $X_t \to B$. Expect that the base manifold B is actually the essential Skeleton.

[Nicaise-Xu-Yu] for any good minimal dlt model $\mathcal{Y}$ of X, the retraction $X^{an} \to Sk(X)$ is an affinoid torus fibration away from the codimension 2 faces of Sk(X).

Affinoid torus fibration is tropicalisation map $(\mathbb{G}_m)^{n\,an} \to \mathbb{R}^n$.

Theorem [Li] an syZ fibration exists on a large region $U_t \subset X_t$ if a non-archimedean conjecture holds. Thus the non-Archimedean picture has concrete consequences on the 'standard' SYZ picture. What's the non-archimedean conjecture? Given a model $(\mathcal{Y}, \mathcal{L})$ of (X, L), there is a distinguished function ϕ on X^{an} which is a solution to the non-Archimedean Monge-Ampere equation. Then the conjecture says that there is a model $(\mathcal{Y}, \mathcal{L})$ such that $\phi = \phi \circ \rho_{\mathcal{Y}}$ over the interior of the maximal faces of Sk(X).